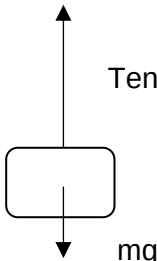


8	(a)	Since <u>the spring constant k and the mass m are both constant</u> , The <u>acceleration of the mass is proportional to the displacement from the equilibrium position</u> and the <u>negative sign</u> indicates that the <u>acceleration and displacement are in opposite directions / acceleration is towards the equilibrium</u> . Hence the oscillation is simple harmonic.		B1 B1
	(b)	(i)	$\omega^2 = \frac{k}{m}$ $(2\pi f)^2 = \frac{35000}{0.020}$ $f = 210.5 = 210 \text{ Hz}$	C1 M1 A1
		(ii)	$F_{\text{max}} = ma_{\text{max}}$ $= m\omega^2 x_0$ $= (0.020)(2\pi(210))^2 (2 \times 10^{-4})$ $= 6.96 = 7.0 \text{ N}$	M1 A1
		(iii)	The maximum tension in the spring occurs when the magnet is at the bottom of the oscillation  $F_{\text{net}} = T - mg$ $T = F_{\text{net}} + mg$ $T = 7.0 + 0.020 \times 9.81$ $= 7.2 \text{ N}$ Zero marks if the weight of the mass was neglected.	M1 A1
		(iv)	$E_{k \text{ max}} = \frac{1}{2} m\omega^2 x_0^2$ $= (0.5)(0.020)(2\pi(210))^2 (2 \times 10^{-4})^2$ $= 7.0 \times 10^{-4} \text{ J}$	M1 A1

	(v)	An inverted parabola for kinetic energy [B1] An upward curve to the right for elastic potential energy [B1] A straight line with negative gradient to the right for gravitational potential energy [B1]	
(c)	(i)	The frequency of the alternating current is <u>210 Hz</u>. This is because when the <u>external driving frequency</u> is equal to the <u>natural frequency</u> of the magnet and spring, <u>resonance occurs</u>, and maximum power transfer to the oscillation. The oscillation will have the <u>largest amplitude</u> and the speaker will produce the loudest sound.	B1 B1
	(ii)	The paper cone and magnet experience air resistance / resistive forces when oscillating which produces <u>light damping</u>. <i>(Any answers which refer to the electromagnetic induction will not be accepted as the circuit is open)</i> Thus without the external periodic force, the <u>amplitude of the oscillation will decay gradually with time</u>.	B1 B1
	(iii)	When the mass of the loudspeaker increases, the <u>natural frequency of the system decreases</u>. Therefore, the (driving) <u>frequency at which the peak amplitude occurs will decrease</u>.	B1 B1
	(iv)	Different speakers of different mass would have <u>different resonant frequencies</u>. or The <u>heaviest speakers would have lower resonant frequencies</u> and the <u>lightest speakers would have higher resonant frequencies</u>.	B1

			or This will allow the loudspeaker box to <u>better produce sounds at different frequencies</u> for broadcasting music or voices. (Any of the above)	